Grade 10 – Term 1: C9 Classroom Activity

All payoffs and regrets are in thousands of dollars. EMV uses the stated summary values. Apply each filter only to current survivors; keep summary values (including regret) fixed. Maximax ranks Max highest first; Maximin ranks Min highest first; Minimax Regret ranks Regret lowest first; EMV/Bayes ranks EMV highest first.

1. City logistics policy.

(a) Complete the Filtering Table. EMV: keep best 4; then Maximin: keep best 2; then Minimax Regret: select best.

Summary

Alt.	EMV	Max	Min	Regret
B_1	70	111	48	99
B_2	62	66	54	96
B_3	84	126	30	72
B_4	86	150	30	111
B_5	79	141	39	111
B_6	90	102	66	84

Filtering

Step	Rule	Survivors	Eliminated
0	Initial	B_1, B_2, B_3, B_4, B_5, B_6	—
1	EMV: best 4		
2	Maximin: best 2		
3	Regret: best 1		

(b) Conceptual Understanding. Why can B_3, whose Max Regret is 72, not reenter the final comparison?

2. Regional distributor expansion.

(a) Complete the Filtering Table. Maximax: keep best 3; then EMV: select best.

Summary

Alt.	EMV	Max	Min	Regret
X_1	26	42	18	108
X_2	88	126	18	87
X_3	85	147	48	66
X_4	83	150	33	93
X_5	76	120	45	105
X_6	104	111	96	39

Filtering

Step	Rule	Survivors	Eliminated
0	Initial	X_1, X_2, X_3, X_4, X_5, X_6	—
1	Maximax: best 3		
2	EMV: best 1		

(b) Conceptual Understanding. Why is X_6, with the highest original EMV of 104, eliminated before EMV is applied?

3. Trade finance guarantees.

(a) Complete the Filtering Table. Maximin: keep best 5; then Minimax Regret: keep best 4; then EMV: keep best 2; then Maximax: select best.

Summary

Alt.	EMV	Max	Min	Regret
G_1	101	141	30	72
G_2	94	150	39	108
G_3	117	147	96	42
G_4	70	78	66	81
G_5	83	135	48	102
G_6	52	102	21	126

Filtering

Step	Rule	Survivors	Eliminated
0	Initial	G_1, G_2, G_3, G_4, G_5, G_6	—
1	Maximin: best 5		
2	Regret: best 4		
3	EMV: best 2		
4	Maximax: best 1		

(b) Conceptual Understanding. Swap the first two criteria: Regret keeps 5, then Maximin keeps 4; leave the last two filters unchanged. Does the final choice change? Show the survivors.

4. Insurance portfolio mixes.

(a) Complete the Filtering Table. Minimax Regret: keep best 4; then Maximax: keep best 2; then Maximin: select best.

Summary

Alt.	EMV	Max	Min	Regret
Y_1	51	69	21	129
Y_2	99	135	30	120
Y_3	117	150	63	72
Y_4	124	150	75	60
Y_5	122	135	114	36
Y_6	68	111	24	111

Filtering

Step	Rule	Survivors	Eliminated
0	Initial	Y_1, Y_2, Y_3, Y_4, Y_5, Y_6	—
1	Regret: best 4		
2	Maximax: best 2		
3	Maximin: best 1		

(b) Conceptual Understanding. If Step 2 kept the best 3 by Maximax instead of 2, would the final choice change? Use the new survivors to justify your answer.

5. Solar project investment.

(a) Complete the Filtering Table. Maximax: keep best 5; then Maximin: keep best 3; then Minimax Regret: select best.

Summary

Alt.	EMV	Max	Min	Regret
S_1	78	87	69	72
S_2	112	132	81	57
S_3	105	138	60	90
S_4	77	102	63	72
S_5	86	129	60	90
S_6	116	150	63	75

Filtering

Step	Rule	Survivors	Eliminated
0	Initial	S_1, S_2, S_3, S_4, S_5, S_6	—
1	Maximax: best 5		
2	Maximin: best 3		
3	Regret: best 1		

(b) Conceptual Understanding. Among the three alternatives entering Step 3, compare the choice by Minimax Regret with the choice by Maximax. What risk attitude does each choice represent?

6. Retail inventory contracts.

(a) Complete the Filtering Table. Maximin: keep best 4; then EMV: select best.

Summary

Alt.	EMV	Max	Min	Regret
R_1	85	93	72	54
R_2	37	66	18	120
R_3	96	138	69	78
R_4	53	99	27	114
R_5	100	138	30	69
R_6	83	147	51	87

Filtering

Step	Rule	Survivors	Eliminated
0	Initial	R_1, R_2, R_3, R_4, R_5, R_6	—
1	Maximin: best 4		
2	EMV: best 1		

(b) Conceptual Understanding. If the final EMV filter were replaced by Minimax Regret, would the selected alternative change? Compare the relevant values.

Solutions

Parentheses show the statistic used at each filtering step; each elimination is new. All summary values remain fixed. No cutoff or final ties occur.

1. City logistics policy

(a)

Step	Rule	Survivors	Eliminated
0	Initial	$\{B_1, B_2, B_3, B_4, B_5, B_6\}$	—
1	EMV: best 4	$B_6(90)$, $B_4(86)$, $B_3(84)$, $B_5(79)$	$B_1(70)$, $B_2(62)$
2	Maximin: best 2	$B_6(66)$, $B_5(39)$	$B_4(30)$, $B_3(30)$
3	Regret: best 1	**Selected:** $B_6(84)$	$B_5(111)$

(b) B_3 is eliminated at Step 2: its minimum 30 is below the cutoff 39. Only B_6 and B_5 reach Step 3. Although $72 < 84$, eliminated alternatives cannot reenter.

2. Regional distributor expansion

(a)

Step	Rule	Survivors	Eliminated
0	Initial	$\{X_1, X_2, X_3, X_4, X_5, X_6\}$	—
1	Maximax: best 3	$X_4(150)$, $X_3(147)$, $X_2(126)$	$X_5(120)$, $X_6(111)$, $X_1(42)$
2	EMV: best 1	**Selected:** $X_2(88)$	$X_3(85)$, $X_4(83)$

(b) X_6 has maximum 111, below the Step 1 cutoff 126. The surviving EMVs are 83, 85, and 88, so X_2 wins with 88 despite X_6 having EMV 104.

3. Trade finance guarantees

(a)

Step	Rule	Survivors	Eliminated
0	Initial	$\{G_1, G_2, G_3, G_4, G_5, G_6\}$	—
1	Maximin: best 5	$G_3(96)$, $G_4(66)$, $G_5(48)$, $G_2(39)$, $G_1(30)$	$G_6(21)$
2	Regret: best 4	$G_3(42)$, $G_1(72)$, $G_4(81)$, $G_5(102)$	$G_2(108)$
3	EMV: best 2	$G_3(117)$, $G_1(101)$	$G_5(83)$, $G_4(70)$
4	Maximax: best 1	**Selected:** $G_3(147)$	$G_1(141)$

(b) Yes. The swapped filters retain G_3, G_1, G_4, G_5, G_2 by regret, then G_3, G_4, G_5, G_2 by minimum. EMV retains $G_3(117), G_2(94)$; Maximax selects $G_2(150)$ over $G_3(147)$. The original choice is G_3.

4. Insurance portfolio mixes
(a)

Step	Rule	Survivors	Eliminated
0	Initial	$\{Y_1, Y_2, Y_3, Y_4, Y_5, Y_6\}$	—
1	Regret: best 4	$Y_5(36)$, $Y_4(60)$, $Y_3(72)$, $Y_6(111)$	$Y_2(120)$, $Y_1(129)$
2	Maximax: best 2	$Y_4(150)$, $Y_3(150)$	$Y_5(135)$, $Y_6(111)$
3	Maximin: best 1	**Selected:** $Y_4(75)$	$Y_3(63)$

(b) Yes. Step 2 would retain $Y_4(150), Y_3(150), Y_5(135)$. Maximin would select $Y_5(114)$ over $Y_4(75)$ and $Y_3(63)$, replacing the original choice Y_4. The tie at 150 is inside the retained set, not at a cutoff.

5. Solar project investment
(a)

Step	Rule	Survivors	Eliminated
0	Initial	$\{S_1, S_2, S_3, S_4, S_5, S_6\}$	—
1	Maximax: best 5	$S_6(150)$, $S_3(138)$, $S_2(132)$, $S_5(129)$, $S_4(102)$	$S_1(87)$
2	Maximin: best 3	$S_2(81)$, $S_6(63)$, $S_4(63)$	$S_3(60)$, $S_5(60)$
3	Regret: best 1	**Selected:** $S_2(57)$	$S_4(72)$, $S_6(75)$

(b) Minimax Regret selects $S_2(57)$ over $S_4(72), S_6(75)$, limiting the worst missed opportunity. Maximax selects $S_6(150)$ over $S_2(132), S_4(102)$, emphasizing the largest possible payoff. These risk attitudes lead to different choices.

6. Retail inventory contracts
(a)

Step	Rule	Survivors	Eliminated
0	Initial	$\{R_1, R_2, R_3, R_4, R_5, R_6\}$	—
1	Maximin: best 4	$R_1(72)$, $R_3(69)$, $R_6(51)$, $R_5(30)$	$R_4(27)$, $R_2(18)$
2	EMV: best 1	**Selected:** $R_5(100)$	$R_3(96)$, $R_1(85)$, $R_6(83)$

(b) Yes. Among R_1, R_3, R_6, R_5, EMV selects $R_5(100)$ over 85, 96, and 83. Regret instead selects $R_1(54)$ over $R_5(69), R_3(78), R_6(87)$.